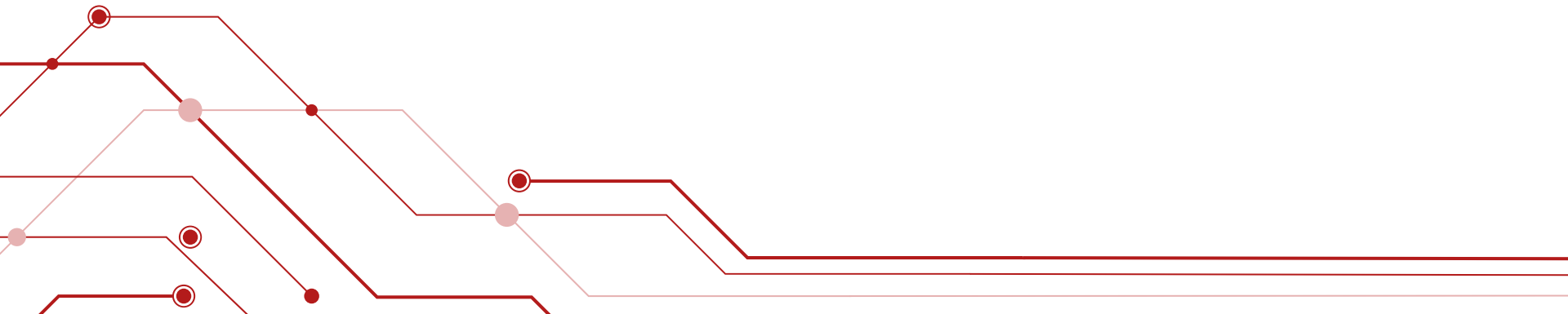


"Adiabatic" Flash ADC

The first (maybe?) of its kind



Introduction to Adiabatic Logic

Adiabatic logic → **Reversible computation**

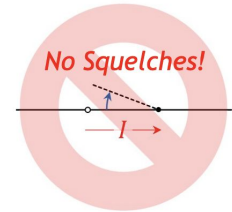
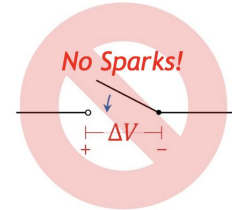
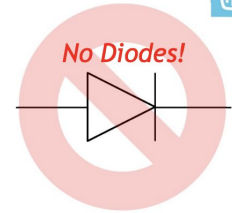
Several **rules** must be followed to allow circuit to be **reversible**

1. Never turn on a transistor when it has a **$V_{DS} > 0$**
2. Never turn off a transistor when there is an **$I_{DS} > 0$**
3. Never use diodes

Slower than pure CMOS, but can **reduce power draw**

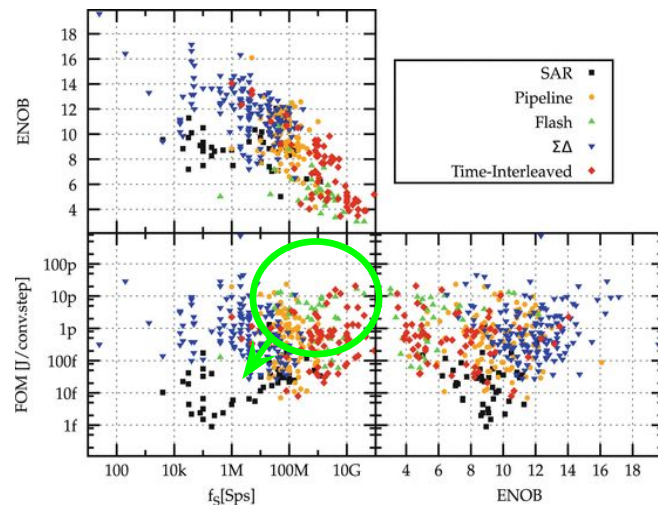
True adiabatic logic requires superconductors

Requires a resonator that can accept and provide power, can, and likely should be, off chip



Rationale for Adiabatic Flash

- Flash ADCs:
 - Power hungry
 - High speed
 - Simple
 - Adiabatic Logic:
 - Low power
 - Moderate speed
 - Higher complexity
- Complement each other!**

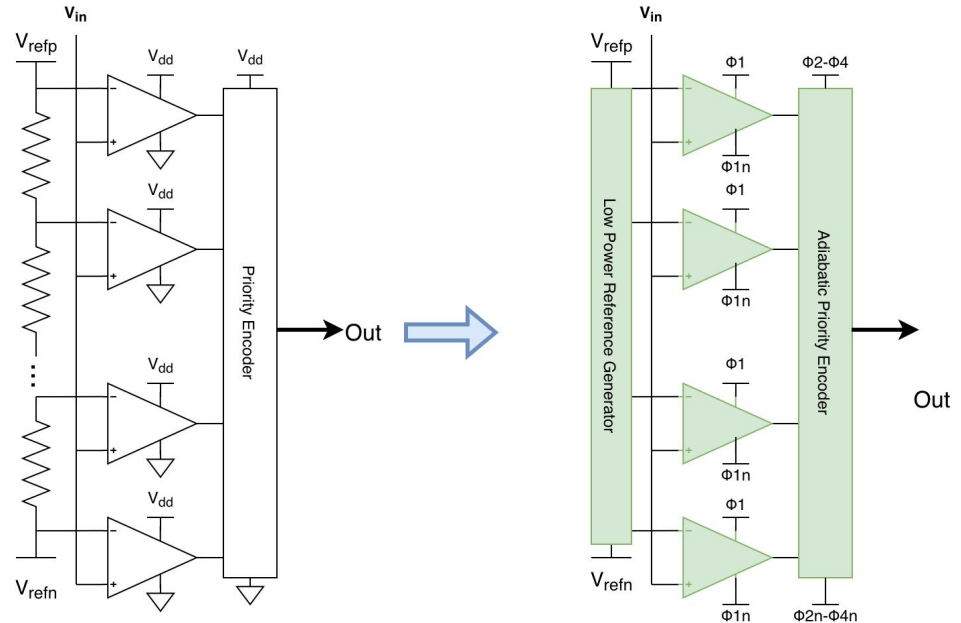


Summary of ADCs published in ISSCC and VLSIC from 2000 to 2015 [B. Murmann]

Goal: Pushing Flash ADCs to low power regime!

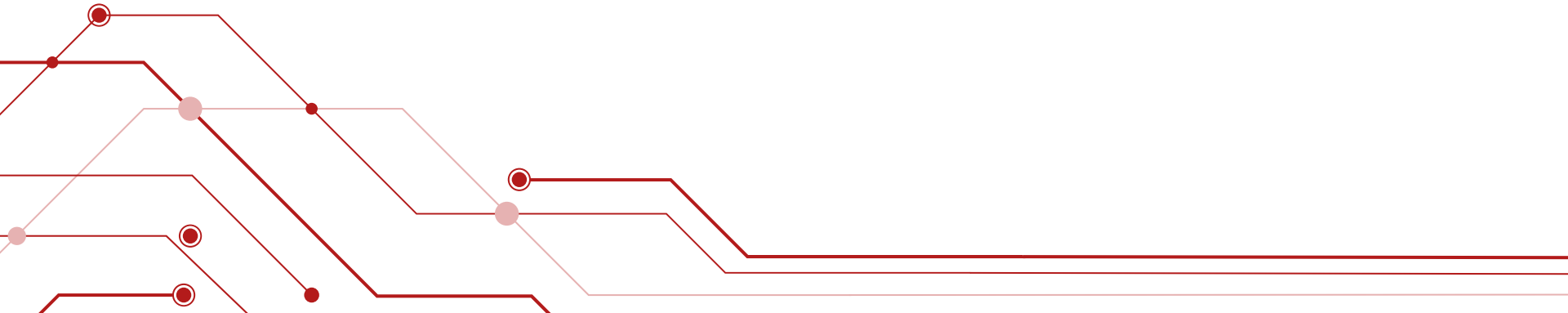
What Makes Flash ADCs Power Hungry?

- **Power Consumers:**
 - Resistor Divider
 - $2^n - 1$ Comparators
 - Digital Encoder
- **Alternatives:**
 - Resonant Cap Divider
 - Low power comparator
 - Adiabatic digital logic



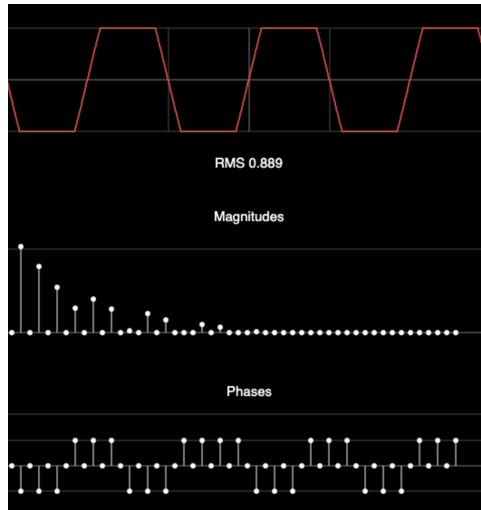


Reference Generation

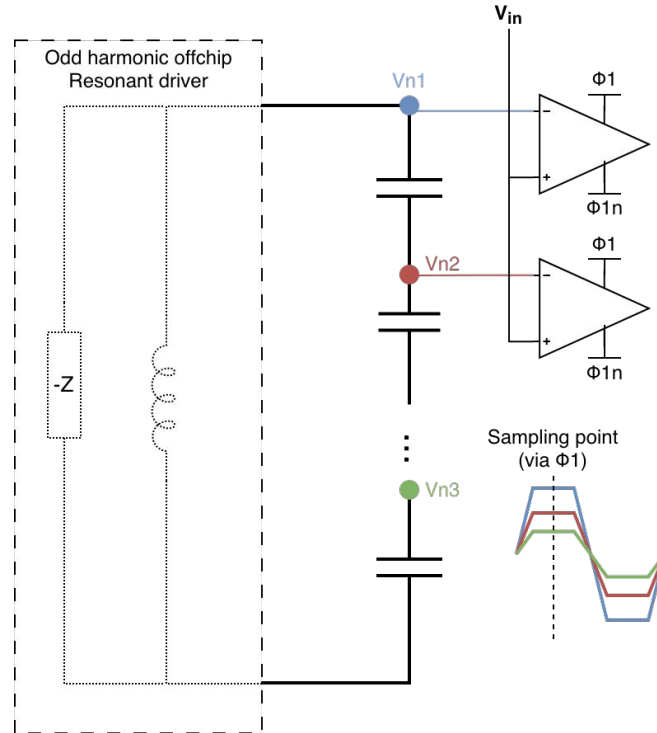


Capacitive Voltage Divider Operation

- Needs “flat part” for extended sampling
- Odd harmonics



Falstad.com/fourier



**So, all is
well?
Not
quite...**

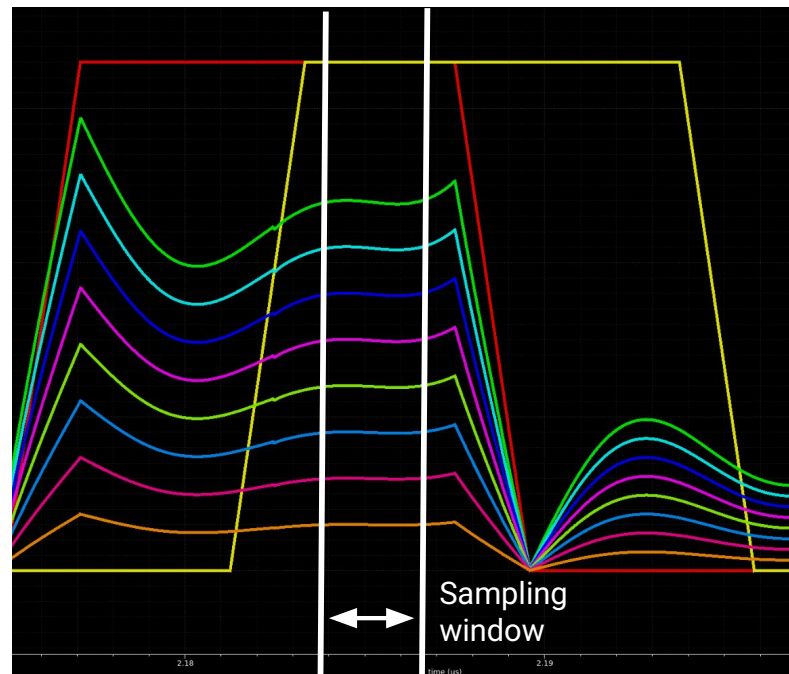
Harmonic Enhanced Cap Divider

Long term charge buildup on capacitors

Sampling window not well defined

Harmonic tuning complex, voltage errors are introduced

A 50k Ω resistor divider draws $\sim 8\mu\text{W}$, comparators draw $\sim 50\mu\text{W}$!



Resistor Divider

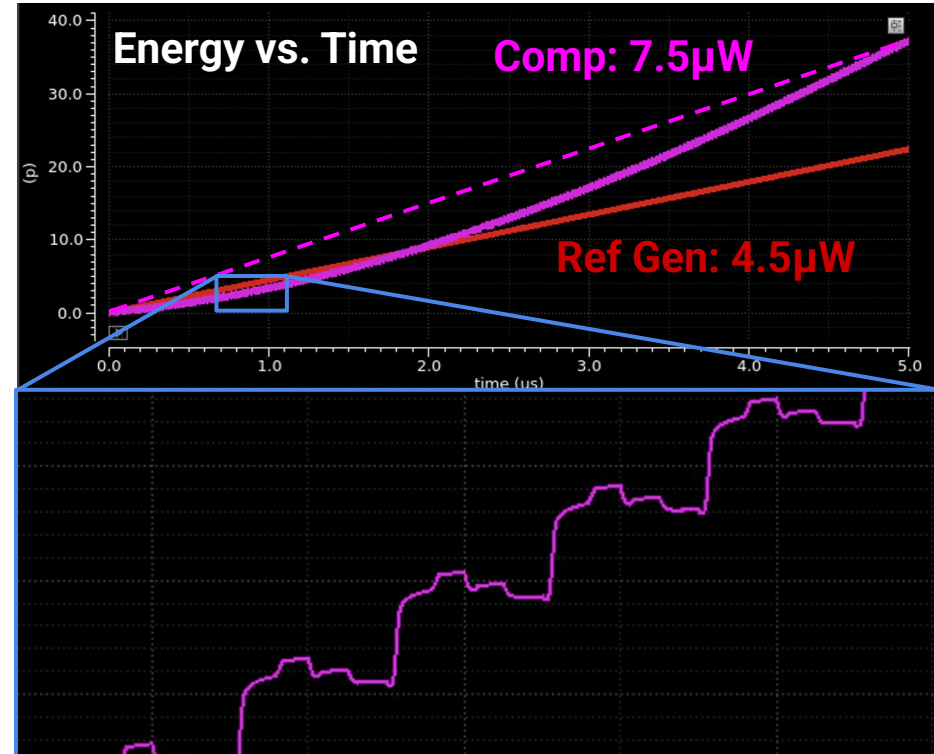
Small power draw compared to 7
adiabatic comparators

Greatly reduced design complexity

No issues with voltage error

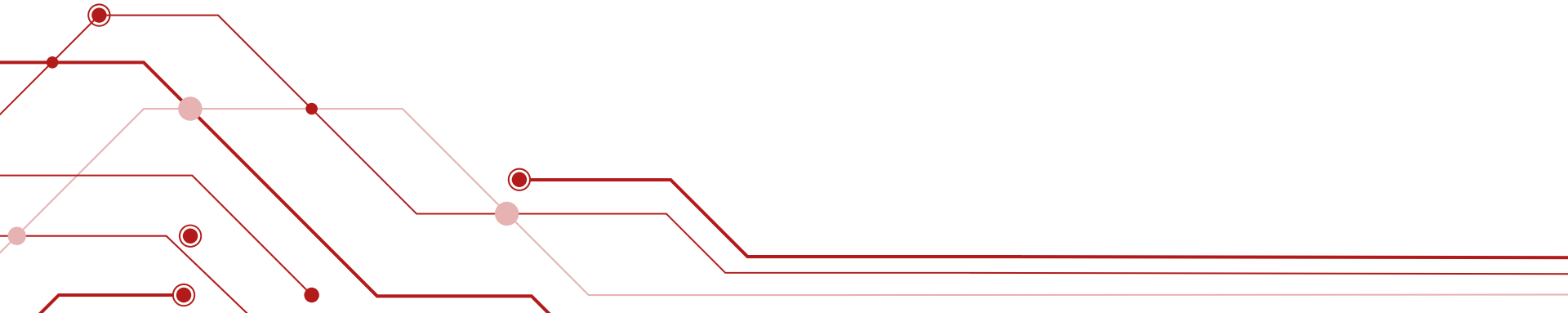
Can mitigate kickback,
decoupling caps (MOSCAs)

Careful layout (common centroid,
dummies, etc.)



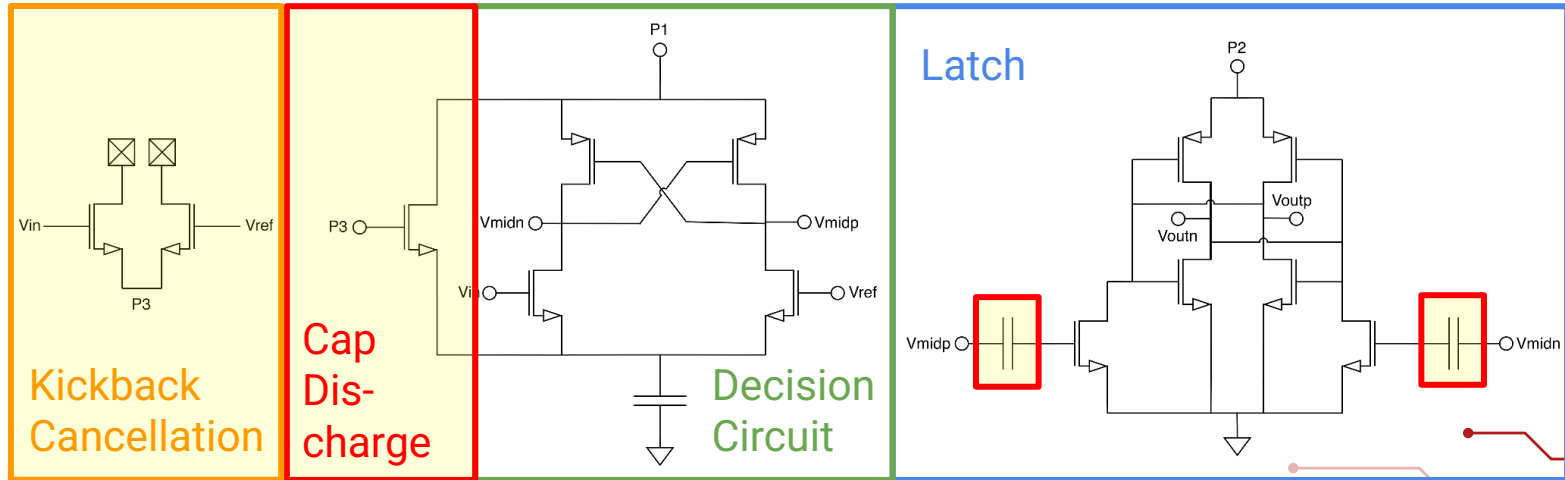


"Adiabatic" Comparator



How to Build An Adiabatic Comparator

- Adiabatic analog circuits are an active area of research
 - Research by Leo Filippini, and Baris Taskin at Drexel, **40fJ/conversion**
- Have to be very careful with sizing, won't work below V_{th} (use LVT-ish)
- Kestrel double tail: Simulated almost **1pJ/conversion @ 40MHz**, 25x higher



Considerations

3.3V device to discharge 1st stage capacitance, coupling caps for biasing

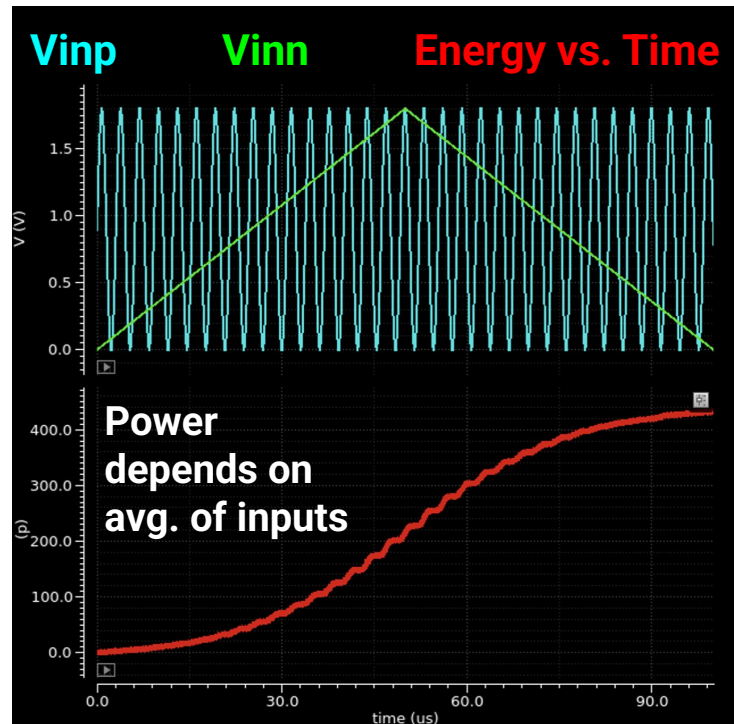
Kickback minimization, cancellation devices

Lowering offset, common centroid, guard rings, symmetric routing

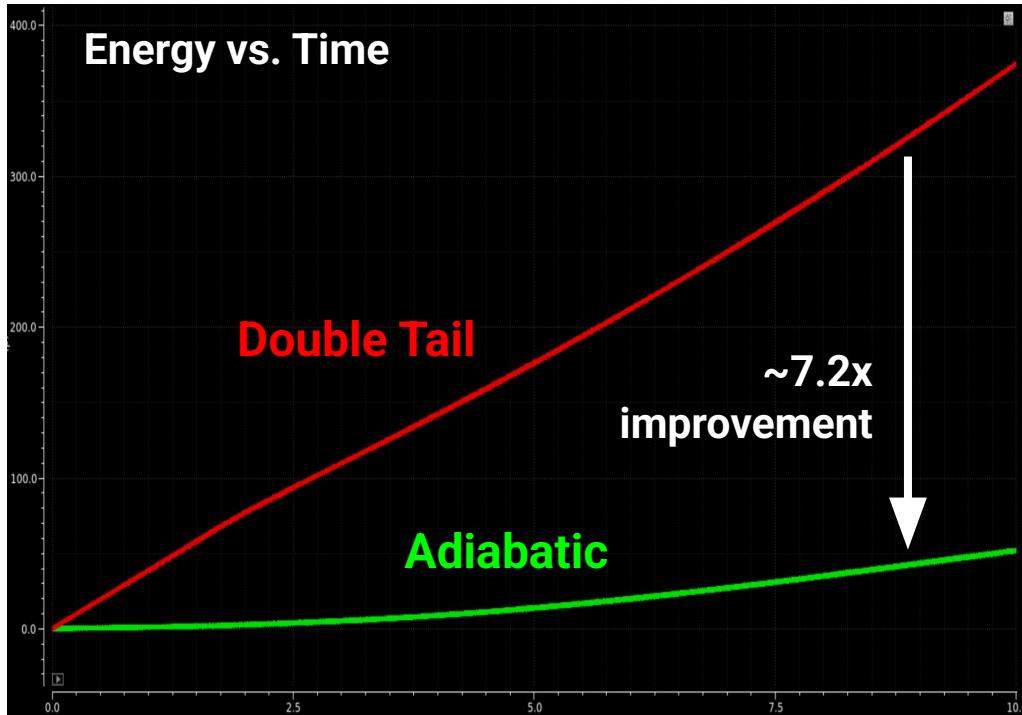
Must work properly from V_{low} to V_{high}

Tried modifications to reduce power, minimal success

Output has to be synchronized with encoder



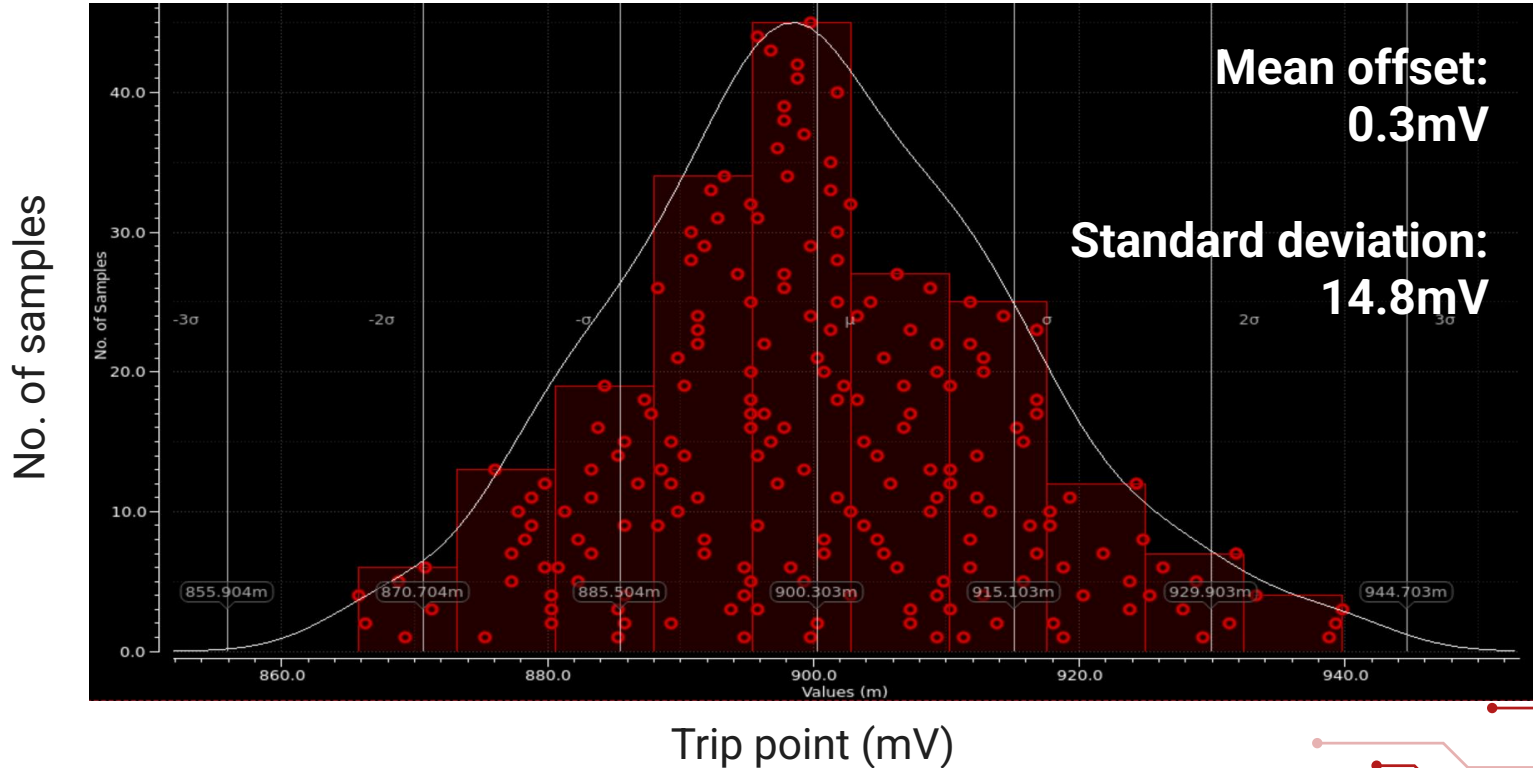
Power Draw



SP25 Double Tail Comparator:
37.5 μ W

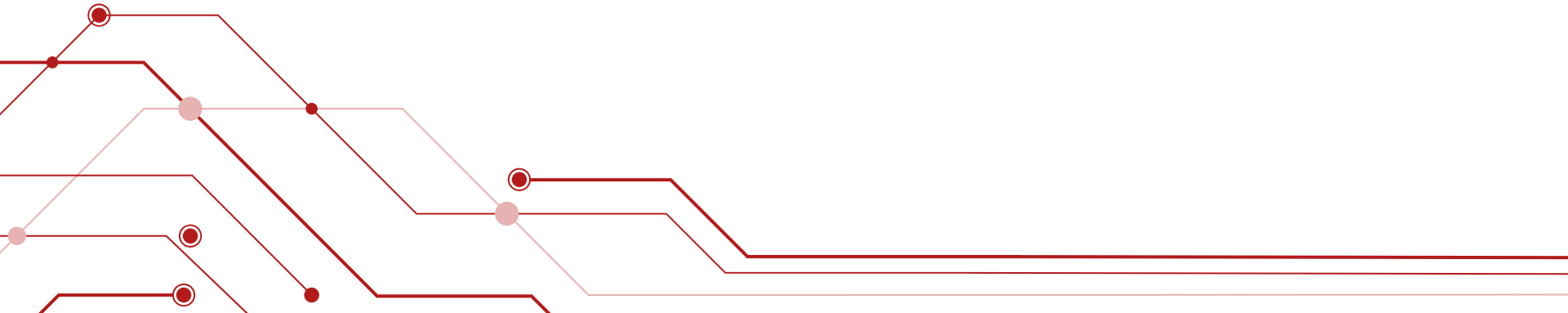
SP26 Adiabatic Comparator:
5.2 μ W

Monte Carlo Simulation





SCRL Priority Encoder



Priority Encoder Purpose

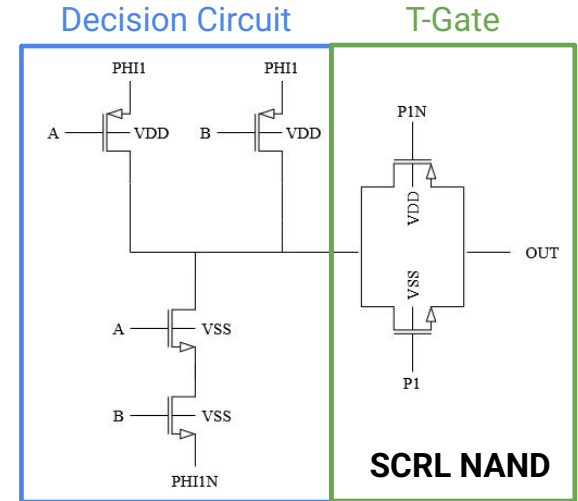
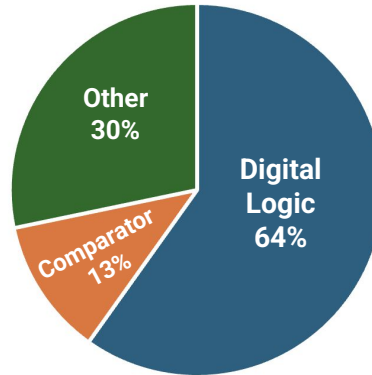
Goal: Take Thermometer Code Comparator Outputs and Turn It Into Binary 3-Bit Output

Input Synchronized (PC-wise) with Comparators

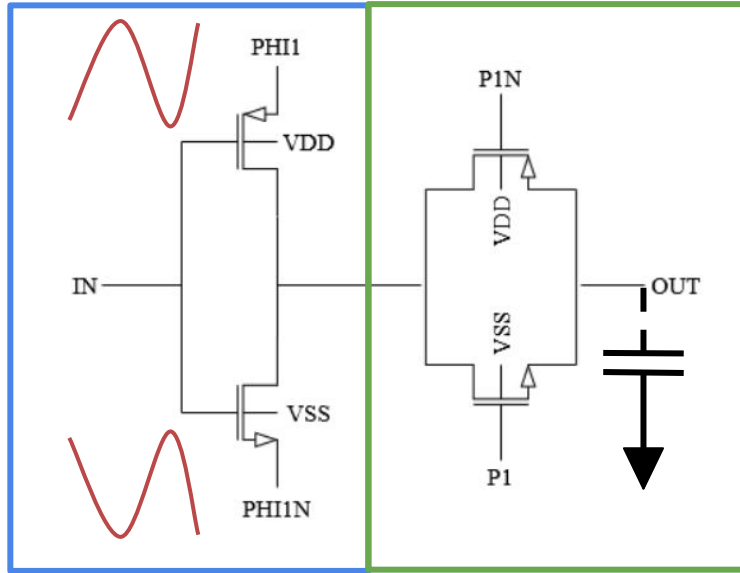
Utilize SCRL (Split-level charge recovery logic) adiabatic logic family

Minimum possible power

Kestrel SAR ADC Power Distribution Chart

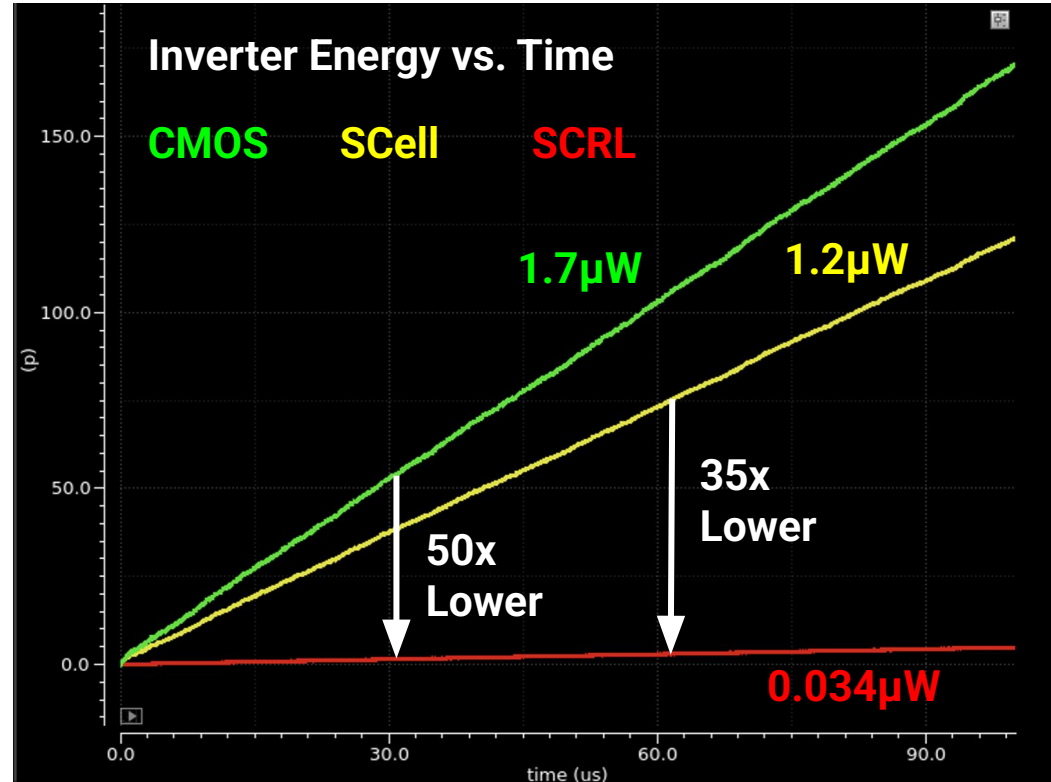


SCRL Standard Cells

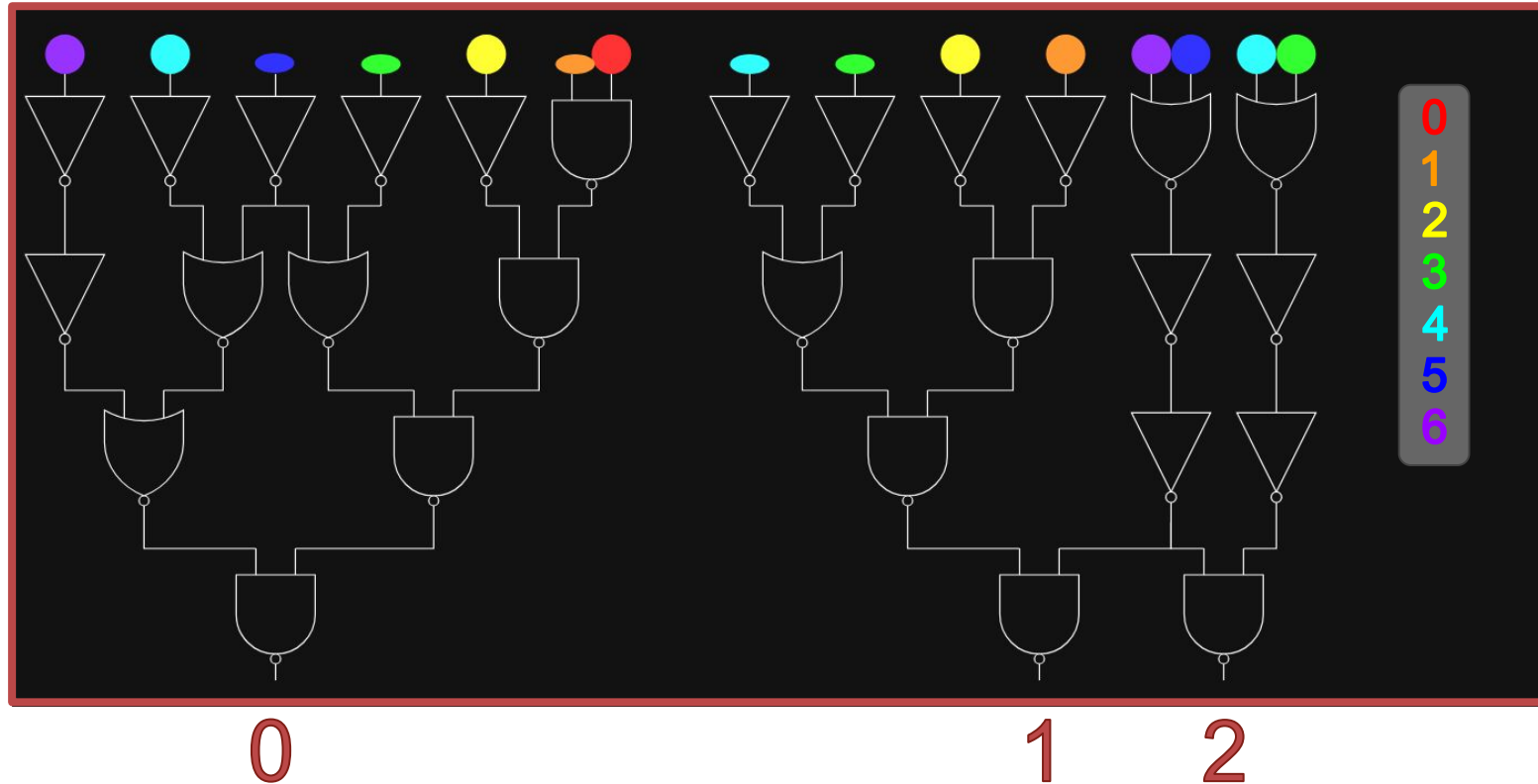


Decision Circuit

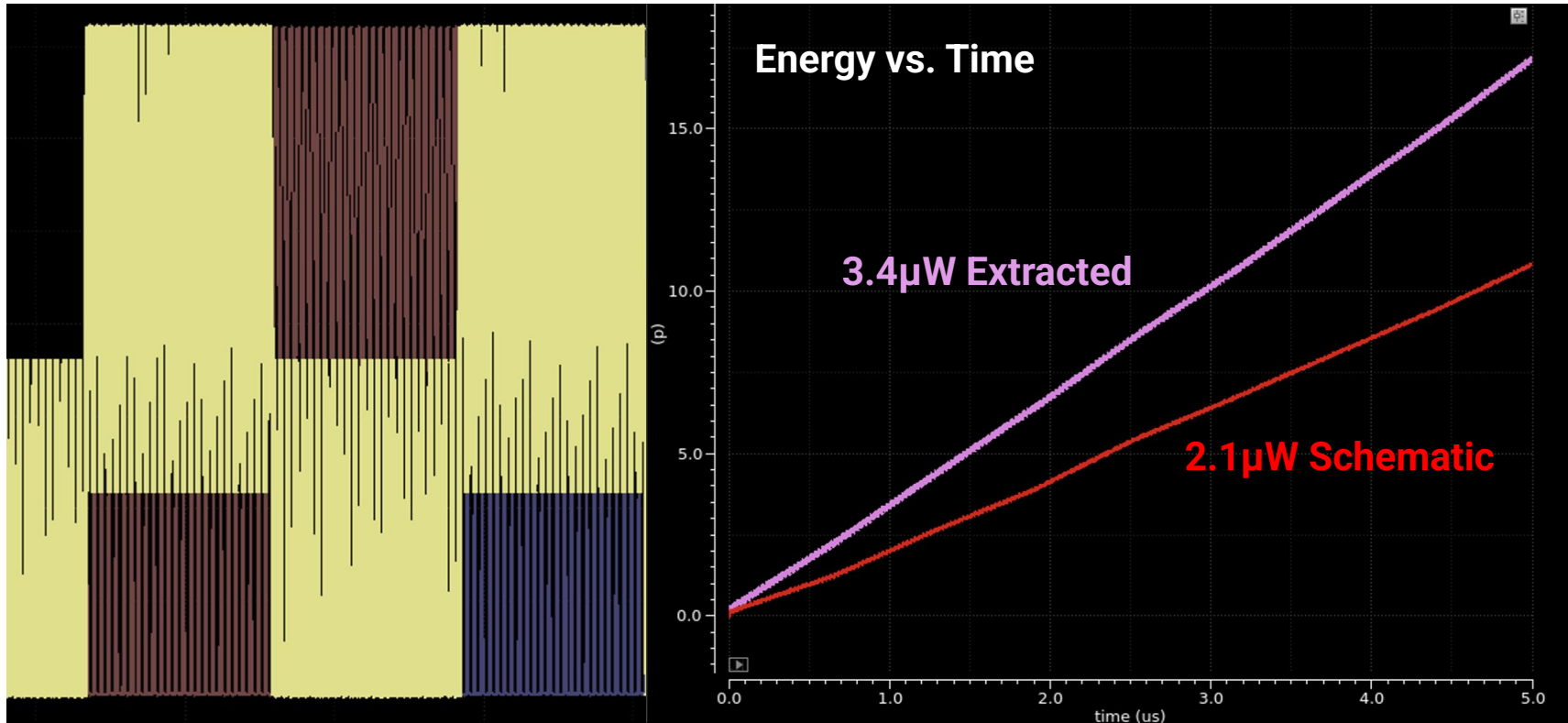
T-Gate

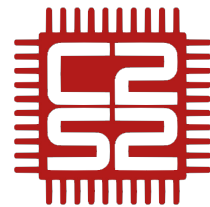


Encoder Schematic

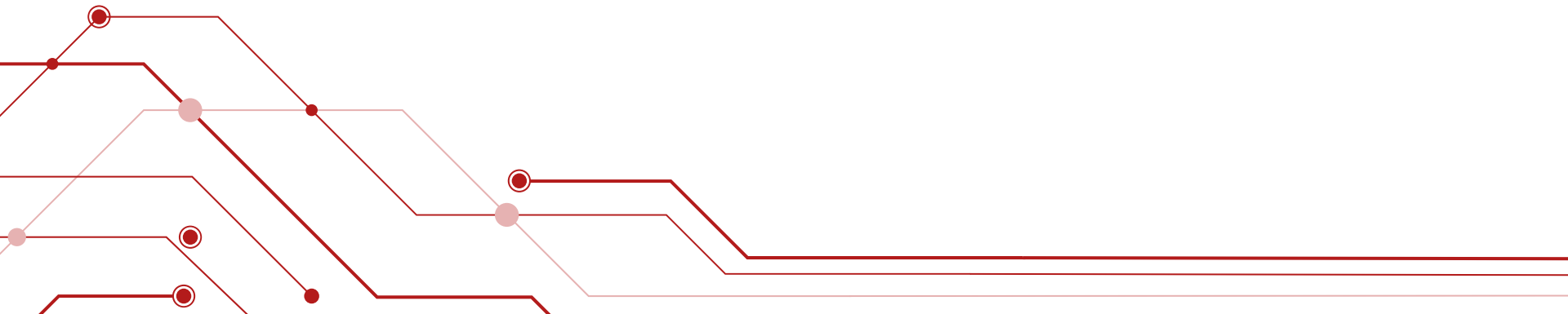


Encoder Performance



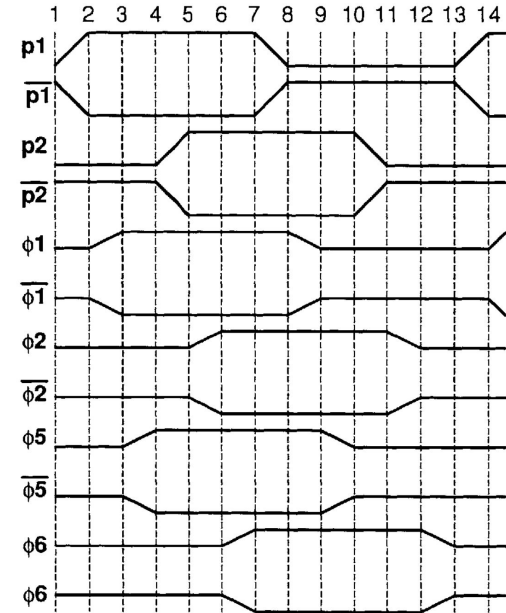


Power Clock Signal Gen



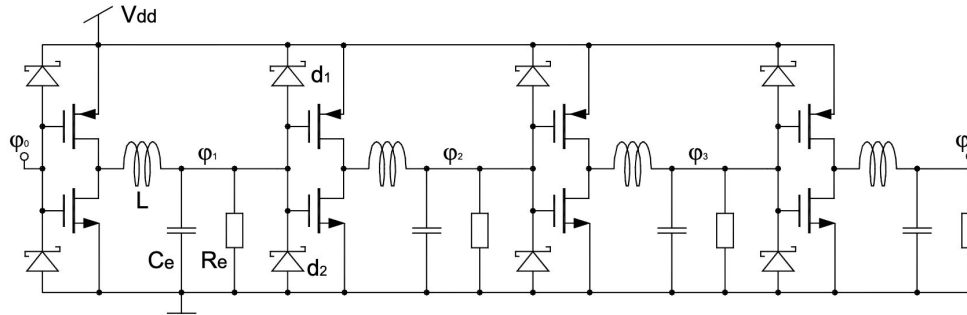
Desired SCRL Waveforms

- Staggered trapezoidal signals to do series logical operations
 - Turn on and off before decision circuit
 - Make adiabatic and keep valid output for next stage
- 16 “phase stages”, ϕ for decision circuit, p for T-gates



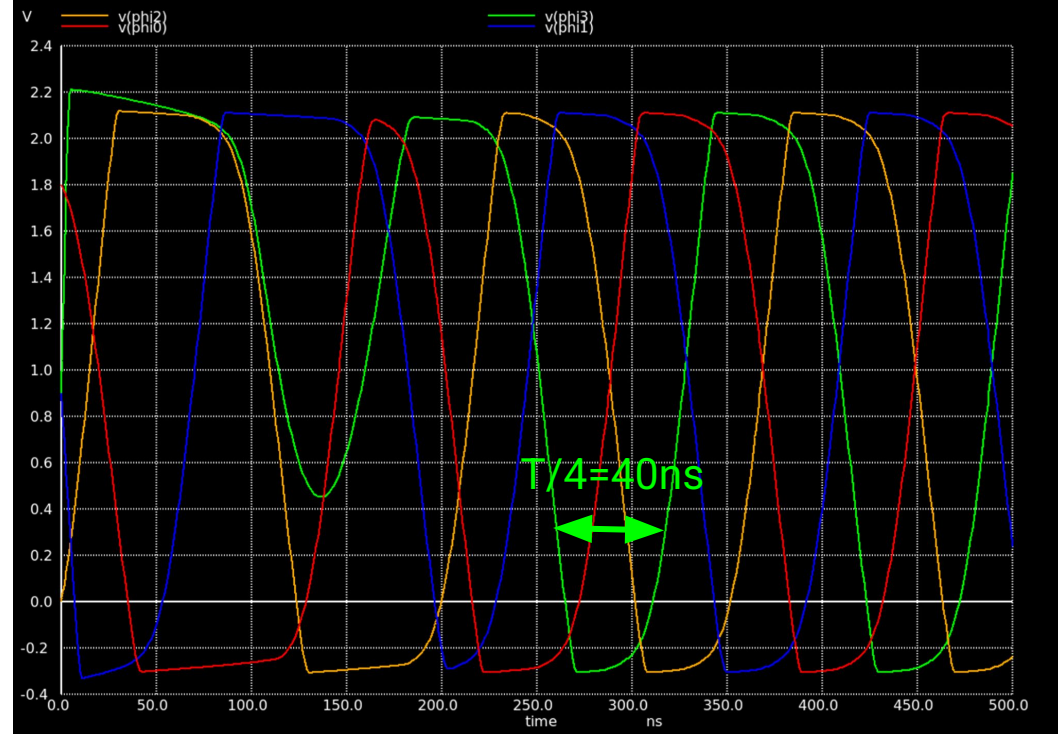
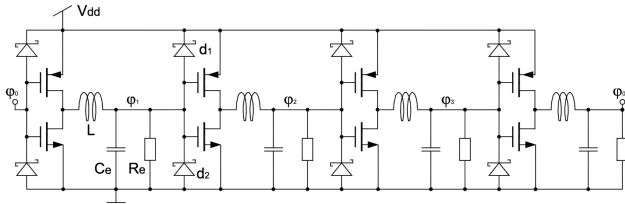
4 Phase Clock Generator

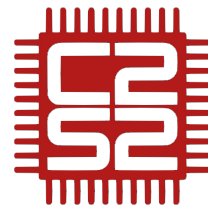
- Single clock generates both the comparator timing and the adiabatic reference ramp
- Energy stores in reference capacitances is recovered into the LC tank each cycle (power isn't dissipated!)
 - Savings grow with 2^N comparators



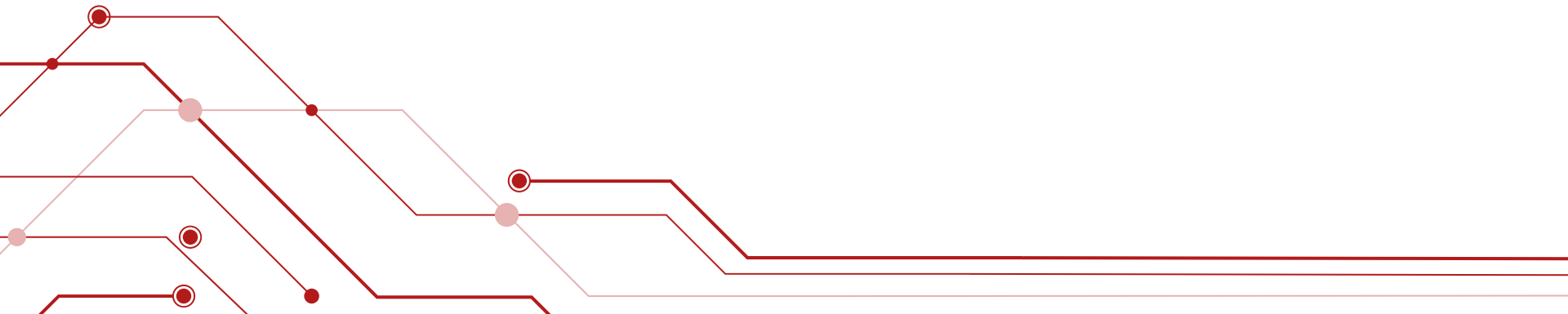
4 Phase Clock Generator

- Simulation reproduces trapezoidal waveforms in paper
- Amplitude asymmetry across all 4 phases: 0.002 mV
- Phase accuracy: <1% error

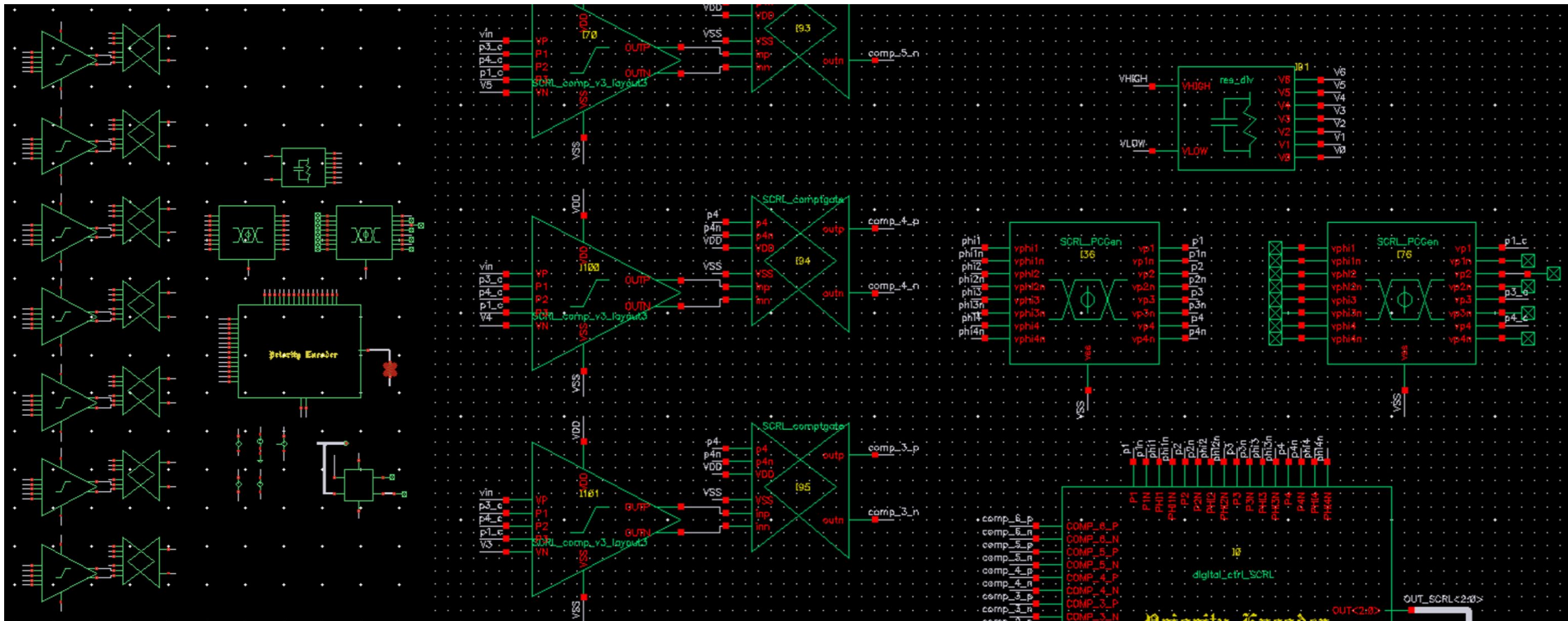




Top Level Integration



Schematic



Testing in Progress

Total power draw of 80 μ W

30x lower than SAR ADC

However, 2^5 less voltage levels

Final layout changes, extracted top level simulations soon

